



Reg.No.:

310914822038

JERUSALEM COLLEGE OF ENGINEERING, CHENNAI- 600100
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Academic Year 2025-2026 (Even Semester)

Assessment- II

Fourth Semester

Common to

B.E CSE, B.Tech CS&BS, B.Tech IT, B.Tech AI&DS, B.E CSE (CS), B.E CSE (AIML)

JCS2401 – DATABASE MANAGEMENT SYSTEMS

(Regulation 2023)

Time:08:30a.m.- 10:00a.m.

Date:07.03.2026

Max.Marks:50

PART A (5×2=10marks)

Answer all the Questions

S.No.	Questions	Course Outcome	Bloom's Taxonomy level
1.	Mention the use of mirroring	CO4	K1
2.	Define Primary index and Secondary Index	CO4	K1
3.	Define OODBMS.	CO5	K1
4.	List the components of mobile database	CO5	K1
5.	What are different isolation levels in database?	CO3	K1

PART B (2×16=32 marks)

6(a)	Illustrate the various types of Hashing, the techniques and issues for each with appropriate examples. (16marks)	CO4	K2
(OR)			
6(b)	List and explain with example the Equivalence rules for Relational algebra expressions. (16 marks)	CO4	K2
7(a)	Explain in detail about XML databases and DTD. (16 marks)	CO5	K2
(OR)			
7(b)	Discuss in detail about (i) Temporal databases (8 marks) , (ii) Spatial Databases (8 marks)	CO5	K2

PART C (1×8=8 marks)

8(a)	Explain in detail about deadlock detection and prevention. (8 marks)	CO3	K2
(OR)			
8(b)	Explain the process of transaction recovery, and save points. (8 marks)	CO3	K2